



CORPORATE STRATEGY

Session 6: Fixed and Variable Costs

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How could a large company fail?

Toys"R"Us. They had \$11B in revenue in 2004. They filed for bankruptcy in 2017. The proximate cause was a \$5.3B LBO debt load but there were other reasons.

According to you what can explain the fall of **Toys"R"Us**?



Types of Costs - Core definitions

Dimension	Definition	Strategic question	Typical cost items
Fixed costs	Costs that do not vary with volume in the short term	What is our breakeven risk and downside exposure?	
Variable Costs	Costs that increase with activity or units sold	What is our marginal profitability per unit?	
Direct costs	Costs clearly attributable to a specific product, service, or client	Can we scale or discontinue this product?	
Indirect costs	Shared or overhead costs supporting multiple activities	How do we allocate without distorting decisions?	

Types of Costs - Core definitions

Dimension	Definition	Strategic question	Typical cost items
Fixed costs	Costs that do not vary with volume in the short term	What is our breakeven risk and downside exposure?	Office rent, HQ staff salaries, core IT systems, depreciation, licenses
Variable costs	Costs that increase with activity or units sold	What is our marginal profitability per unit?	Raw materials, packaging, logistics per unit, payment fees, commissions
Direct costs	Costs clearly attributable to a specific product, service, or client	Can we scale or discontinue this product?	Product COGS, dedicated labor, product-specific marketing, cloud usage per product
Indirect costs	Shared or overhead costs supporting multiple activities	How do we allocate without distorting decisions?	Management salaries, HR, finance, brand marketing, shared infrastructure

Evolution of costs - Cost behavior matters more than cost labels

Cost behavior	Description	Typical examples
Linear variable	Proportional to volume	Raw materials, transaction fees
Step-fixed	Fixed within a capacity range, jumps when capacity increases	Warehouses, servers
Semi-variable	Fixed base + variable component	Salaries + commissions, energy

Cost structure archetypes

Archetype	Cost structure	Typical sectors
Asset-heavy	High fixed / low variable	Industry, airlines
Asset-light	Low fixed / high variable	Marketplaces, services
Software-like	High fixed / near-zero variable	SaaS, platforms
Labor-driven	Semi-variable	Consulting

Operating leverage: the strategic bet behind cost structure

A company with high fixed costs and low variable costs has high **operating leverage**
⇒ small revenue changes produce amplified profit changes.

$$\text{Operating leverage} = \text{Contribution margin} / \text{EBIT}$$

NETFLIX


ArcelorMittal

 airbnb

Make or Buy: Transaction Cost Theory

Why do firms exist at all?

Ronald Coase (1937) answered it: because using markets has costs. When the cost of transacting through the market exceeds the cost of organizing internally, the firm internalizes.

	Make (internalize)	Buy (outsource)
Cost type	Coordination costs (management, overhead, capex)	Transaction costs (search, contracting, enforcement, dependency)
Risk	Higher fixed exposure	Higher variable exposure + supplier power

Oliver Williamson (1975, Nobel 2009) operationalized Coase. Three factors increase transaction costs and push toward internalization:

1. Asset specificity : How unique is this asset to your business? If the asset (skill, process, component) is highly specific to your product, no supplier can deliver it efficiently. You must own it.

2. Uncertainty : How hard is it to write a complete contract? If the activity involves rapid change, complex specs, or proprietary knowledge, contracts are always incomplete. You lose control if you outsource.

3. Frequency : How often does this transaction recur? High-frequency, recurring transactions justify the fixed cost of building internal capability. One-off needs do not.

Decision rule: The higher the score on these 3 factors, the stronger the case to Make.

Two real examples

Apple outsources iPhone manufacturing to Foxconn (Buy) but internalizes chip design (Make).

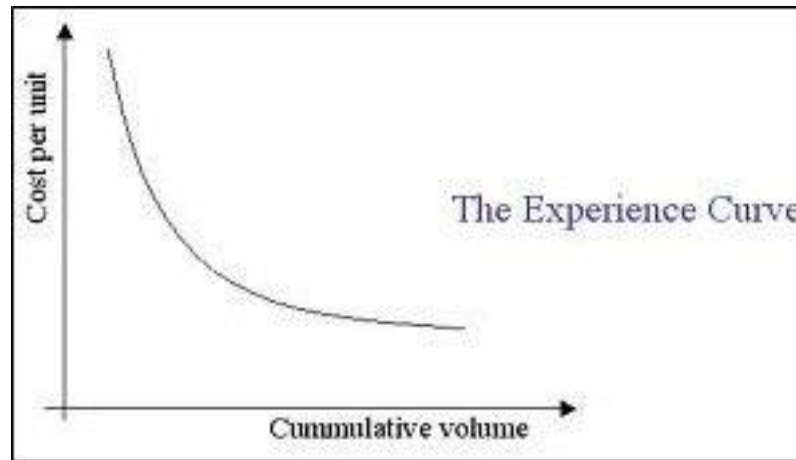
- **Why outsource manufacturing?** Foxconn operates at 1M+ workers on dedicated Apple lines. The transaction is high-frequency but low in specificity for Apple's competitive position. Foxconn's scale produces a cost advantage Apple cannot replicate.
- **Why internalize chip design?** The M-series and A-series chips are highly asset-specific (built for Apple's OS and product integration), involve deep uncertainty (no complete contract can capture Apple's roadmap needs), and are high-frequency decisions.

Tesla internalized battery production via the Gigafactory Nevada (\$5B investment, 2016).

- **Asset specificity:** battery cost represented ~40% of vehicle cost, and Tesla's cell chemistry is tailored to its own BMS and vehicle architecture
- **Uncertainty:** Panasonic as sole supplier created unacceptable contract incompleteness. Tesla had no control over cost trajectory or production pace
- **Frequency:** every vehicle requires thousands of cells -- maximum recurrence

Cost as competitive moat: the experience curve

(BCG, 1966): Each time cumulative production doubles, unit costs fall by a predictable percentage (typically 15-25%) due to learning effects, process optimization, and scale.



Solar panels

Cost per watt fell from ~\$76 in 1977 to ~\$0.20 in 2023. Every doubling of cumulative installed capacity reduced cost by ~20%. This is why early market share in solar was worth more than current margin.

Samsung semiconductors

By investing ahead of demand in DRAM capacity (late 1990s), Samsung rode the experience curve faster than Micron and Hynix. By 2000, Samsung had ~35% cost advantage per chip. Strategy: accept losses short-term to win cost leadership permanently.

Business Case (1/2)

You are advising a **premium DNVB (Digitally Native Vertical Brand)** operating in the home & lifestyle segment. Management now asks a **simple but critical question**: *At this price point, can we make money and what must we optimize first?*

You are tasked with analyzing the **cost structure** and drawing **strategic conclusions** in choosing the best scenario; low case: **500 units**, base case: **1,200 units**, high case: **2,000 units**

Item	Amount (€ / unit)
Selling price (ex-VAT)	120 €
Manufacturing (COGS)	42 €
Packaging	4 €
Logistics & fulfillment	9 €
Payment & platform fees	3 €
Team (founders + 4 FTE)	45,000 €
Marketing & brand (fixed part)	25,000 €
Tech stack & tools	8,000 €
General & admin (office, legal, accounting)	12,000 €

Q1. Contribution margin

- What is the **contribution margin per unit (€)**?
- What is the **contribution margin (%)**?

Q2. Breakeven analysis

- What is the **monthly breakeven volume**?
- In which volume scenario(s) is the company profitable?

Q3. Strategic lever

- Which **single cost lever** has the **highest strategic impact**? Price, Variable costs, Fixed costs, Volume

Business Case (2/2)

1.

Question	Calculation	Explanation
What is the contribution margin per unit?	Price – Variable costs → 120 € – 58 € = 62 €	The contribution margin represents the cash generated by each unit sold to cover fixed costs and then generate profit. As long as this margin is positive, selling more units improves profitability <i>after breakeven</i> .
What is the contribution margin (%)?	Contribution margin / Price → 62 / 120 ≈ 52%	The margin rate indicates how much of each euro of revenue is available to absorb fixed costs. A high percentage provides strategic flexibility (pricing, promotions, shocks).

2. Breakeven volume= 90 000 / 62€ ≈ 1,450 units/month

3. Now test each lever with a +5% improvement (disciplined comparison):

Lever	Change	New CM/unit	New breakeven	Impact
Price +5%	120 → 126€	68€	90,000 / 68 = 1,324 units	-128 units (-8.8%)
Variable cost -5%	58 → 55.1€	64.9€	90,000 / 64.9 = 1,387 units	-65 units (-4.5%)
Fixed cost -5%	90k → 85.5k€	62€	85,500 / 62 = 1,379 units	-73 units (-5.0%)
Volume +5%	1,452 → 1,525 units	no change to breakeven	P&L improves, breakeven unchanged	different mechanic

Cost management has an impact on the company

Strategic decision	Cost question	Methodological lens	Decision logic
Pricing change	What happens to contribution margin?	Contribution margin analysis	A price change is acceptable only if contribution margin remains positive and improves breakeven or strategic positioning
Growth	Does scale improve profitability?	Cost behavior & breakeven analysis	Growth is value-creating only if contribution margin > 0 and fixed costs are absorbed faster than volume increases
Outsourcing	Fixed → variable trade-off	Cost variability & risk analysis	Outsourcing is justified if flexibility and risk reduction outweigh higher variable unit costs
Portfolio pruning	Which products destroy margin?	Contribution margin by product	Products with negative or structurally weak margins should be redesigned or exited, regardless of revenue

Cost management has an impact on the strategy

Make (in-house)	Buy (outsource)
Higher fixed costs	Higher variable costs
Better control & learning	More flexibility
Economies of scale	Faster entry

In-house	SaaS
High fixed (dev, maintenance)	Variable / subscription-based
Customization	Standardization
Long-term cost advantage	Short-term flexibility

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Business Case - Should the DNVB internalize logistics (Make) or outsource to a partner (Buy) as volumes grow?

Demand is growing but still volatile (Current volume: 1,200 units / month ; Growth ambition: 2,500+ units / month)

. Logistics is currently outsourced to a 3PL. Management is considering internalizing logistics to improve margins.

Scenario Buy

Cost type	Amount
Variable logistics cost	9 € / unit
Fixed logistics cost	0 €
Capacity limit	None

Scenario Make

Cost type	Amount
Variable logistics cost	4 € / unit
Fixed cost (base capacity)	30,000 € / month
Capacity limit	3,000 units / month
If volume > 3,000 => 2 warehouse employees; +10,000 € fixed costs	

Operational reality

- When logistics is **internalized**, delivery times improve
- This improves conversion rate and reduces paid marketing pressure

Estimated impact

- With BUY: marketing fixed cost = **25,000 € / month**
- With MAKE: marketing fixed cost = **22,000 € / month**

Your mission

1. Compute **total monthly logistics + marketing cost**

1. For BUY and MAKE
2. For each volume level (800 units / month; 1,200; 2,500; 3,200; 4,000)

2. Identify:

1. The **economic breakeven** (cost-based)
2. The **operational breakpoint** (capacity-based)

3. Decide:

1. Should the DNVB internalize logistics?
2. If yes: **when?**

Business Case – Should the DNVB internalize logistics (Make) or outsource to a partner (Buy) as volumes grow?

Volume (units/month)	BUY: Total cost (€)	MAKE: Total cost (€)
800	$(9 \times 800) + 25,000 =$ 32,200	$(4 \times 800) + 30,000 +$ $22,000 =$ 55,200
1,200	$(9 \times 1,200) + 25,000$ $=$ 35,800	$(4 \times 1,200) + 30,000 +$ $22,000 =$ 56,800
2,500	$(9 \times 2,500) + 25,000$ $=$ 47,500	$(4 \times 2,500) + 30,000 +$ $22,000 =$ 62,000
3,200	$(9 \times 3,200) + 25,000$ $=$ 53,800	$(4 \times 3,200) + 40,000 +$ $22,000 =$ 74,800
4,000	$(9 \times 4,000) + 25,000$ $=$ 61,000	$(4 \times 4,000) + 40,000 +$ $22,000 =$ 78,000

Element	Value
Fixed cost difference	30,000 €
Variable cost difference	$9 - 4 =$ 5 € / unit
Breakeven volume	$30,000 / 5 =$ 6,000 units